MIT ACADEMY OF ENGINEERING

DSA PRACTCIAL ASSIGNMENT -08

(GRAPH)

NAME:- Nandini Chimandare

PRN:- 202501110108

BRANCH:-CSE(AIML) (B1)

TOPIC:- Implement a program to check whether the given string is palindrome or not using doubly ended Queue.

AIM:-

To implement a C++ program to check whether a given string is a palindromeor notusing a doubly ended queue (Deque).

DESCRIPTION:-

Adeque (DoubleEnded Queue) is a linear data structure in which insertion and deletion can be performed at both the front and rear ends.

In this program, the characters of the given string are inserted into a deque. The first and last characters are compared. If they match, they are removed from both ends, and the process continues until one or no characters remain. If all the characters match, the string is a palindrome; otherwise, it is not.

CODE:-

```
#include <iostream>

#include <deque>

#include <string>

using namespace std;
```

```
int main()
{
    deque<char> dq;
    string str;
    int flag = 1;

    cout << "Enter a string: ";
    cin >> str;

    for (int i = 0; i < str.length(); i++)
    {
        dq.push_back(str[i]);
    }

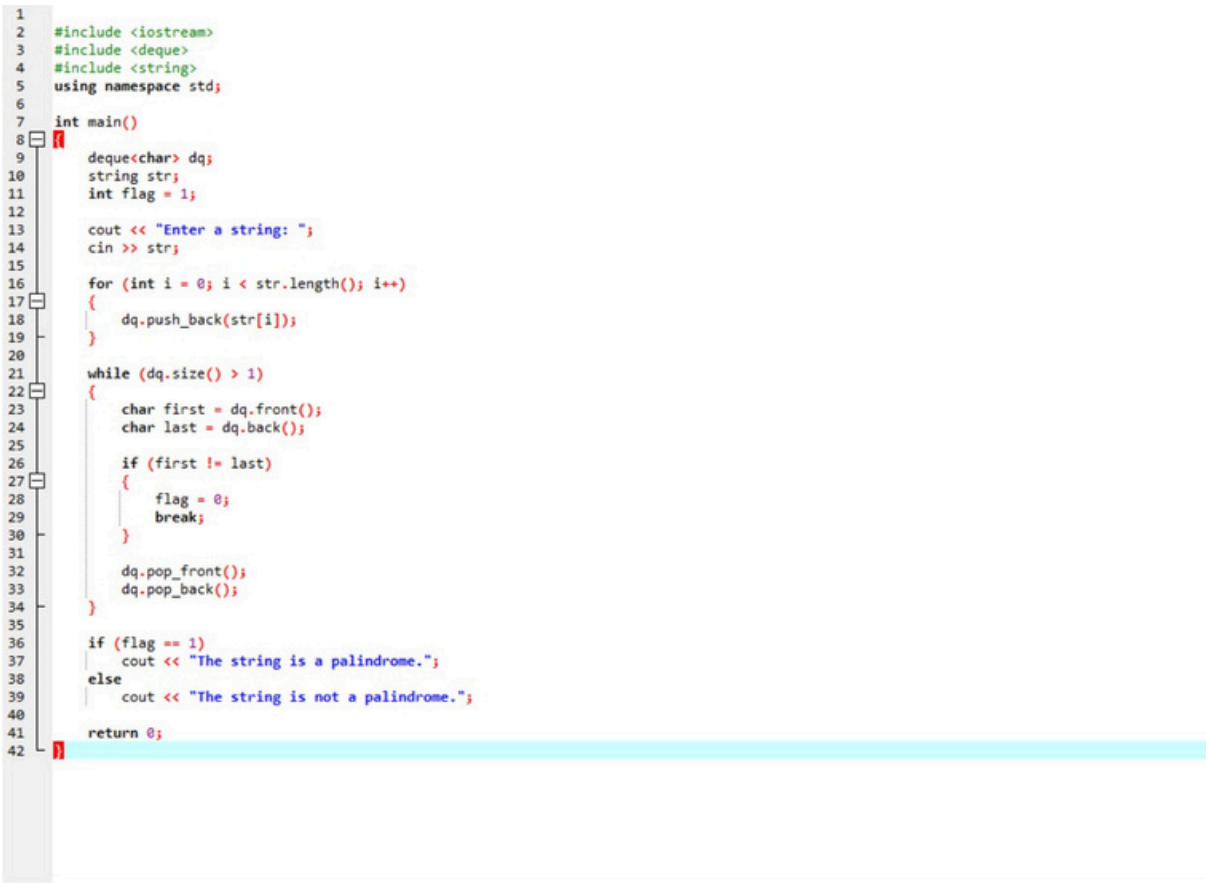
    while (dq.size() > 1)
    {
        char first = dq.front();
        char last = dq.back();

        if (first != last)
        {
            flag = 0;
            break;
        }
```

```
        dq.pop_front();
        dq.pop_back();
    }

    if(flag == 1)
        cout << "The string is a palindrome.";
    else
        cout << "The string is not a palindrome.";

    return 0;
}
```



```
#include <iostream>
#include <deque>
#include <string>
using namespace std;

int main()
{
    deque<char> dq;
    string str;
    int flag = 1;

    cout << "Enter a string: ";
    cin >> str;

    for (int i = 0; i < str.length(); i++)
    {
        dq.push_back(str[i]);
    }

    while (dq.size() > 1)
    {
        char first = dq.front();
        char last = dq.back();

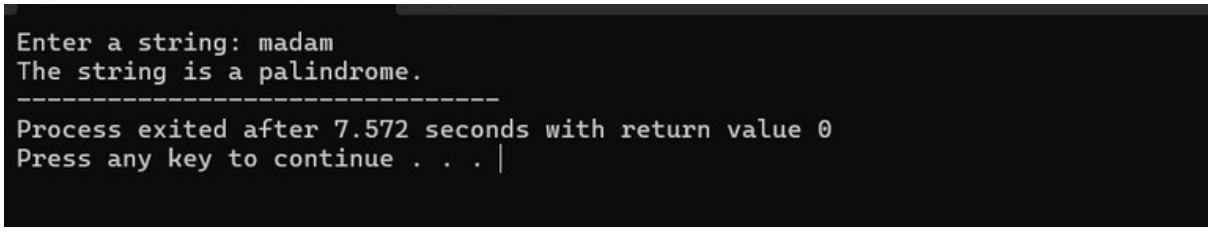
        if (first != last)
        {
            flag = 0;
            break;
        }

        dq.pop_front();
        dq.pop_back();
    }

    if (flag == 1)
        cout << "The string is a palindrome.";
    else
        cout << "The string is not a palindrome.";

    return 0;
}
```

OUTPUT:-



RESULT:-

Theprogram wassuccessfully implemented and executed to check whetherthe givenstring is a palindrome using a deque.

CONCLUSION:-

Thus,theDSA program to check whether a given string is a palindrome using a doubly ended queue was successfully implemented.